

VREE: A 3D Environment to Train Engineers about Post-Earthquake Damages in Building Structure

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ABSTRACT

Using the latest advancements in technology, such as virtual reality (VR), can increase student engagement in classrooms and improve their learning skills in the Science, Technology, Engineering, and Mathematics (STEM) fields. Therefore, now is the time to explore further how VR can be used more effectively to targeting real-life applications in STEM fields. The Virtual Reality Environment for Engineering (VREE) project focuses on developing a training platform for civil engineering students and practitioners to learn about post-earthquake damage assessment of structures. The platform enables trainees to go around a 3D building structure to train and diagnoses different kinds of damages in a post-earthquake scenario.

Index Terms: Applied computing—Education—

1 INTRODUCTION

In the case of teaching engineering concepts in classroom settings, instructors may not have sufficient live examples to demonstrate properly many critical concepts important to understanding real-life applications. For example, due to the scale of structural projects (e.g., buildings, bridges, stadiums, etc.), most Civil Engineering (CE) instructors cannot provide hands-on experiences to students such as manipulating and handling structural elements. Only select universities have the resources to access construction sites for their engineering students or expose them to research facilities capable of testing scaled and full-sized systems. Teaching through traditional approaches (e.g., pictures and videos) may limit students' abilities in understanding properly such concepts, which may result in reduced class engagement.

Kinzie et al. [5] mentioned that not acquiring the desired knowledge, skills and competencies are the main reasons behind low retention and college completion rates in students. Previous studies (e.g., [4]) have described classroom student engagement as a major factor in student success. Therefore, it is time to explore how real world understanding through the use of state-of-the-art technologies, such as VR, will help students in getting the desired knowledge, skills, and competencies as well as increase class engagement by bridging the gap in accessible resources.

Not surprisingly, VR's presence in education is growing [1,3]; it is time to investigate how it can improve learning in the classroom [6] and what kinds of learning activities can effectively be done to increase student personal development and comprehension [7]. Within CE, several studies [2, 8, 9] have initiated work to bring VR into the field ranging from simple structural systems to overviews of processes with results supporting improvements in students' cognition.

Structural failures due to design/material problems amount to only several hundred over the course of a decade [10]. In the case of these real-life engineering failures, access to observe these failures is extremely limited due to the risks of further structural failure/collapse,



Figure 1: The original building used as a proof-of-the-concept.

fall hazards, etc. This leaves the majority of structural education left to using traditional visual aids such as photographs or videos to reinforce motivation for design procedures. With the goal of enhancing students' skills, competencies, and class engagement, we developed a simulated environment to train civil engineering students and practitioners about different kinds of post-earthquake damage to buildings. The resulting VR environment, called **VREE** (Virtual Reality Environment for Engineering), enables CE students to get training on understanding different kinds of structural damages without the risks of going physical there in the building. Also, the environment is created to evaluate students' and professionals' knowledge of structural damages in order to make sure that they have obtained the required skills and competencies. Furthermore, we believe that use of such technologies in the classroom settings would increase students' engagement, which would motivate degree pathways and increase retention/graduation rates.

2 THE USE-CASE SCENARIO

The VREE team was established with computer science and civil engineering faculty and students at San Francisco State University (SFSU), USA. The team collaborates with the Structural Engineer's Association of Northern California (SEAONC¹) to get the required training and educational material in structural damages as well as potential professional users to evaluate the developed environment. As a proof of concept, we used an original building (see Fig. 1, situated near the University of California, Berkeley campus (USA), that stands as a four-story apartment with a ground-level parking garage and three residential levels above. This particular building structure was chosen for its representative nature, mirroring common residences in the Bay Area of California, USA. Its construction details include a masonry and concrete parking lot, timber-framed stories, glass windows facing the atrium, and a brick facade.

3 THE VIRTUAL REALITY ENVIRONMENT FOR ENGINEERING

In the VREE project, first the foundational elements of the targeted building structure in Figure 1 were modeled in Autodesk Revit

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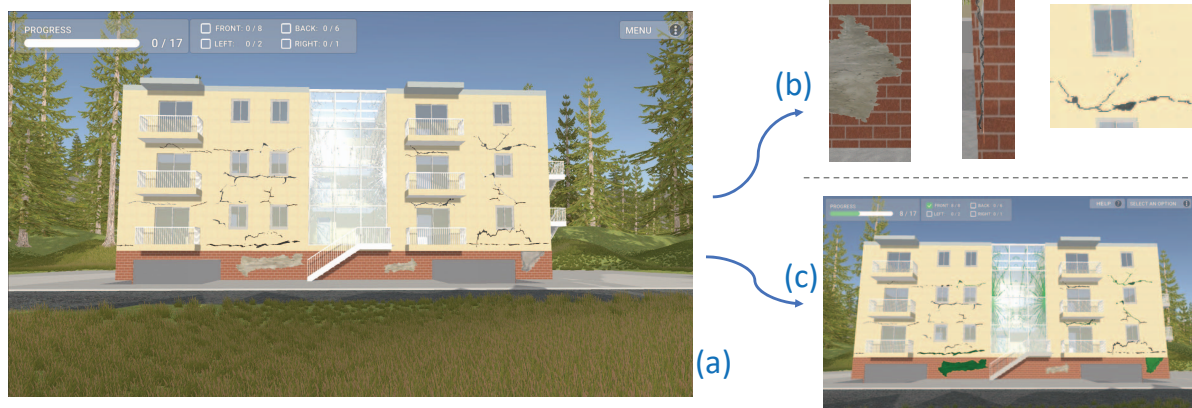


Figure 2: (a) The targeted building in VR-Engine environment with post-earthquake damages, (b) different damages types, (c) correct identified damages are highlighted in green while top-progress bar shows the progress of overall damages recognition.

provide architectural design details. Then we imported it into Autodesk Maya, a 3D computer graphic application, for textual and material assignment. Finally, the building model was integrated into Unity3D Game Engine (see Fig. 2(a)), as the main development platform for developing the proposed 3D VR environment to provide the required structural damages to the targeted building for the training purpose. Unity3D's FBX file format seamlessly supports 3D animation from Autodesk Maya, facilitating smooth integration for modeling, animating, rendering, and compositing. At this stage, we focused primarily on the post-earthquake damage development to the building exterior. Therefore, users are able to go around the whole building structure to look, observe and analyze different kinds of damages. VR-Engine enables the users to navigate around the building through mouse, touchpad or keyboard keys.

We used the structural damage categories based on the training material provided by the SFSU School of Engineering and SEAONC to simulate different kinds of damages, e.g., surface cracks, brick peeling, and glass cracking. We decided to place strategically those structural damages to different parts of the building in order to simulate post-earthquake scenario more realistic. We used decal textures from online textures libraries (e.g., *textures.com* and *freepnglogos.com*) to simulate the desired damages to the building (e.g., see Fig. 2(b)). We also used *Unity Terrain Tool* package to sculpt custom terrains for providing the surrounding environment to the targeted structure and assets from *Unity Asset Store* to enrich the environment elements for giving a more realistic environment feelings.

The VREE environment provides a **Training Mode**, which enables students to learn more about the kinds of damages as well as to evaluate their learned knowledge through finding out those damages in the structure. In this mode, VREE enables the users to identify damages upon clicking and then recognize them based on their categories. Upon selecting a specific damage, it becomes highlighted (see Fig. 2(c)) as well as provide a sound effect based on correct or incorrect identification, to visually and audibly confirm its identification and then users can select the damage category. A dedicated score panel with progress-bar positioned at the top of the environment that shows users damages identification progress (see Fig. 2(c)). This is especially useful for training students, so they know how many damages they are able to recognize correctly.

The current VREE environment enables CE students to learn about post-earthquake damages to the building structure and to evaluate their learned knowledge about it. Currently, the environment is created from the perspective of using it in desktop display settings. We are working on moving the environment towards head-mounted displays. Furthermore, we intend to use it in the classroom setting

with civil engineering students at the School of Engineering, SFSU, in a senior design project course to see the usability of the environment as well as to see how it enhances students' learning and engagement. We also intend to evaluate the environment with professional engineers at SEAONC to see the feasibility of the environment to be used as a training tool for professional engineers.

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